

Mechanical properties and dynamic strain aging behavior of Zr–1.5Nb–0.4Sn–0.2Fe alloy

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Abstract

To elucidate the dynamic strain aging behavior of Zr–1.5Nb–0.4Sn–0.2Fe alloy which was developed as an advanced nuclear fuel cladding material, tensile tests were carried out for the alloy in the range of 25–500 °C at the strain rate of 8.33×10^{-5} and $1.67 \times 10^{-2} \text{ s}^{-1}$. The results showed that the alloy exhibited the characteristics of the dynamic strain aging behavior above 200 °C such as an increase in the strength, work hardening exponent as well as a decrease in the ductility and strain rate sensitivity. An increase of the heat treatment temperature from 470 to 510 °C shifted the temperature range of the dynamic strain aging to a higher temperature range because the initial dislocations causing the dynamic strain aging were annihilated by the heat treatment. The analysis of the activation energy and the diffusion model of the solute atom revealed that the dynamic strain aging in Zr–1.5Nb–0.4Sn–0.2Fe alloy was due to a thermal migration of vacancies generated around the substitutional solute atoms in addition to a diffusion of the interstitial solute atoms.

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1. Introduction

The cladding tube for a nuclear power plant is a reactor core material, which prevents nuclear material from leaking into the coolant as the first safety wall of a nuclear fuel. To select a material suitable for the cladding tube, the reactor system designer should consider some important factors such as the mechanical properties as well as the resistance to an irradiation and corrosion in the reactor core system. Zirconium alloys such as zircaloy have been used since the 1960s as the material because they meet the criteria and have a small cross section for neutron absorption. A low Sn zircaloy-4 alloy containing about 1.3 wt.% Sn has been selected for this material since the 1990s because it has a stronger corrosion resistance than the standard zircaloy-4 alloy, which contains about 1.5 wt.% Sn. Recently, USA, France and Japan have attempted to develop new zirconium alloys for the cladding tube to enhance the fuel economy by lowering the corrosion rate of the fuel rod. The Korea Atomic Energy Research Institute (KAERI) has recently developed some cladding mate-

rials for a high burn-up nuclear fuel to test their performance. The materials are Zr-based multi-component alloys containing a little amount of Sn, Nb, Fe, Cr and Cu to the extent as needed in order to increase the corrosion and creep performance [1–2]. In the range of 200–700 °C, it was reported that the strength of zirconium alloy increased but the ductility decreased due to a dynamic strain aging (DSA) from the mutual interaction between the moving dislocations and the solute atoms [3]. A DSA is referred to as the phenomenon in which an alloy is hardened with a decrease of the ductility when it is deformed with a specific strain rate in the range of a specific temperature. It is recognized as the mutual interaction between the solute atoms and the dislocations, and the interference of impurities in the moving dislocations, etc. [4,5]. In the DSA range, the hardening of an alloy as an aging effect could be observed early after a deformation and even during it. In such a case, it is hard to forecast the properties of a cladding tube at the operating temperature of a power plant. Also a DSA decreases the design margin of the fuel system in the operational environment of a power plant. The data [6] on the change of the mechanical properties of the newly developed alloy for the cladding can be used for the calculation of the design margin. This study was implemented to observe the changes of the mechanical properties of the newly developed

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Zr–1.5Nb–0.4Sn–0.2Fe alloy in the range of 25–500 °C at the strain rate of 8.33×10^{-5} and $1.67 \times 10^{-2} \text{ s}^{-1}$ and to review a DSA mechanism.

2. Experimental procedure

2.1. Test material and specimens

A cladding tube made of a multi-component Zr alloy (Zr–1.5Nb–0.4Sn–0.2Fe) for the test was manufactured by pilgering an ingot which had been prepared with a vacuum arc re-melting. A component analysis after a manufacture revealed that the tube contained 100 ppm carbon, 50 ppm nitrogen and 1200 ppm oxygen as solute atoms. It is known that the corrosion and mechanical properties of the Zr alloy can be sensitively changed with their final heat treatment [1,2]. To identify the changes of the mechanical properties, the tube specimens were given a final heat treatment for 2.5 h at 470 and 510 °C, respectively, before the test. Fig. 1 shows the microstructure of the specimen which was stress-relieved at 470 °C and partially re-crystallized at 510 °C.

2.2. Tensile test

To obtain the data on the mechanical properties and observe the DSA phenomenon, the tensile tests of the tube were carried out at room temperature, 200, 250, 280, 310, 340, 370, 400, 450 and 500 °C after temperature stabilization for 20 min. The tests were implemented in a vacuum environment of 10^{-2} Torr. To evaluate the mechanical properties of the tube, axial tensile test specimens with 25 mm in gage length were prepared as defined in Fig. 2. The deformation of the specimens was measured from the displacement of the cross head. The specimens were deformed with the strain rate of 8.33×10^{-5} and $1.67 \times 10^{-2} \text{ s}^{-1}$. The other details for the tensile test were done according to ASTM B811 and ASTM E8M. After test the fractured surfaces were observed using the SEM. Oxygen content in the specimens was analyzed using LECO TC-136 equipment.

3. Results and discussion

3.1. Stress and strain

Fig. 3 shows the stress–strain curves of the alloy with the different temperatures. With an increase of the temperature, the stress and Young's modulus decreases but the ductility increases except for around 340 °C. Fig. 4 magnifies the early deformation section on the curve in Fig. 3. For the specimens with the final heat treatment at 470 °C, a fine serration on the graph was found in the plastic range when they had been tested at 340 °C with the strain rate of $8.33 \times 10^{-5} \text{ s}^{-1}$. The serration was a type of positive saw tooth, which is recognized as a phenomenon which occurs in alloys when the moving dislocations are fixed due to an active DSA [7]. However, when the specimens were given a final heat treatment at 510 °C a serration was not found in the plastic range but a yield drop was found in the range of 25–200 °C. It appears that why the serration was not observed is related to DSA intensity. The DSA intensity of specimens with the final heat treatment at 470 °C could be higher than that with the final heat treatment at 510 °C because the former was only stress-relieved to have its initial dislocation density which could capture more solute atoms but the latter has a lesser dislocation density to capture as much due to a partial re-crystallization. Since the velocity of the moving dislocation in a Zr alloy is almost equal to the diffusivity of a solute atom such as oxygen at 340 °C [8], it seems that the specimen with the final heat treat-

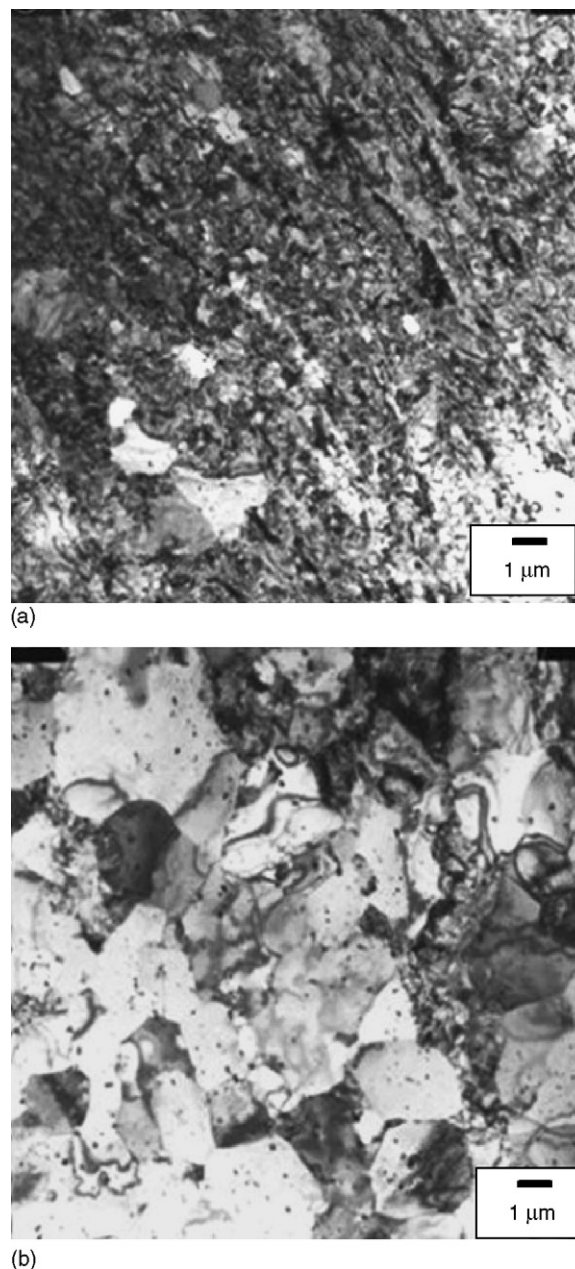


Fig. 1. TEM microstructures of Zr–1.5Nb–0.4Sn–0.2Fe cladding tube heat treated at (a) 470 °C and (b) 510 °C.

ment at 470 °C shows a serration on the stress–strain curve but that with the final heat treatment at 510 °C does not. The yield drops in the specimens with the final heat treatment at 510 °C might be from the release of the dislocations which were fixed to the solute atoms with a higher stress or with the help of a ther-

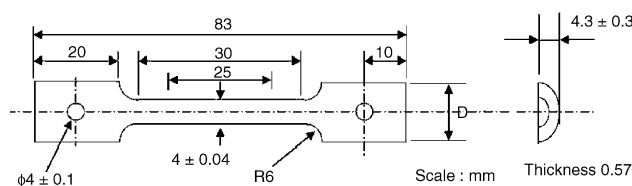


Fig. 2. Tensile specimen of Zr–1.5Nb–0.4Sn–0.2Fe cladding tube.

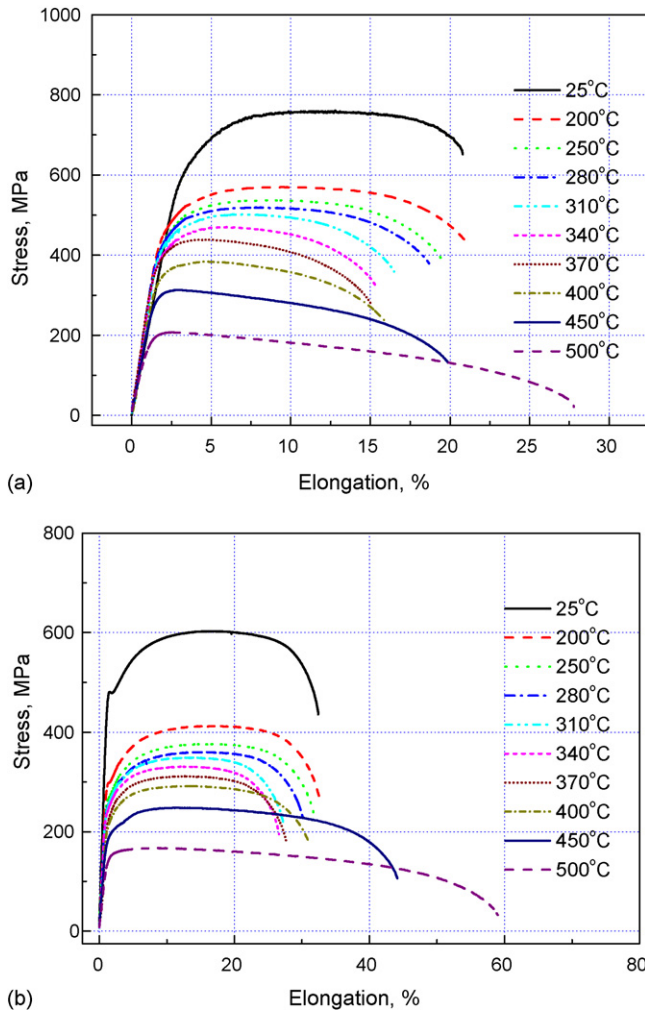


Fig. 3. Stress–strain curves of Zr–1.5Nb–0.4Sn–0.2Fe cladding tube, heat treated at (a) 470 °C and (b) 510 °C, strained at $8.33 \times 10^{-5} \text{ s}^{-1}$.

mal fluctuation at a higher temperature. If the specimens have a large number of dislocations, the dislocations which locate in the neighborhood of sessile dislocation can accommodate an additional deformation without releasing the pinned dislocation. Since the specimens heat-treated at 470 °C have a higher initial dislocation density than those at 510 °C, the effect of yield drop in the specimens seems not so much when compared to those heat-treated at 510 °C.

Fig. 5 shows the ultimate tensile stress (UTS) of the specimens with a change of the temperature. The higher the heat treatment temperature is, the lower the UTS is because the specimens would recrystallize and their initial dislocation density would decrease with an increase of the temperature. In the stress–temperature diagram, the stress of the Zr alloy would decrease with the temperature but the magnitude of the decrease would become lower in the DSA region. It is known that the strength of Zr alloy would decrease with a 0.5 power of the test temperature if it did not have the DSA effect [9]. If a DSA of the specimens would occur a little at room temperature and 500 °C, the intrinsic stress to the temperature can be related with 0.5 power of the test temperature. The relation is

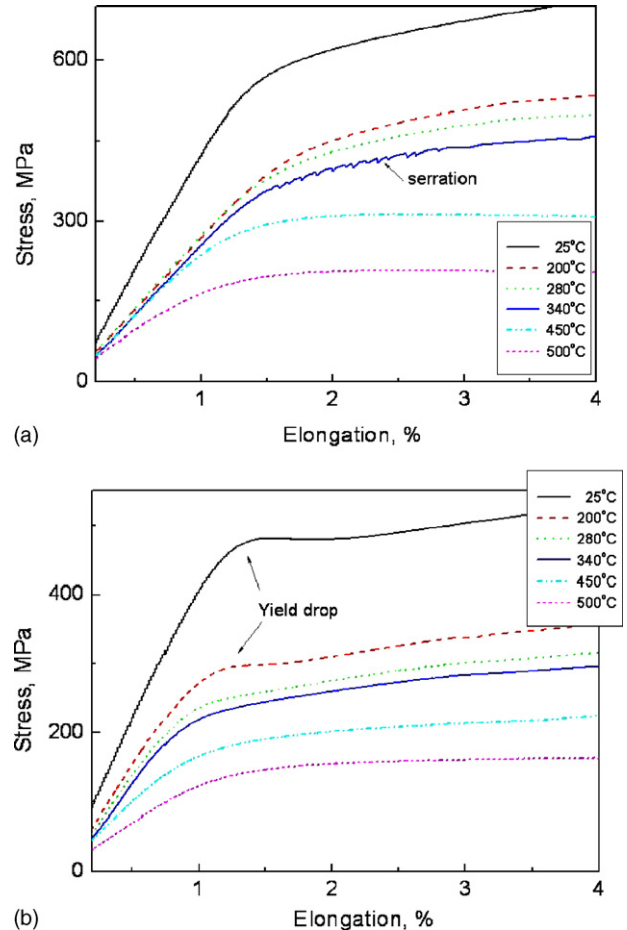


Fig. 4. Magnified stress–strain curves of Fig. 3 for early elongation of Zr–1.5Nb–0.4Sn–0.2Fe cladding tube, heat treated at (a) 470 °C and (b) 510 °C, strained at $8.33 \times 10^{-5} \text{ s}^{-1}$.

indicated as a straight line in Fig. 5. The line value shows the theoretical strength which decreases with an increase of the temperature and it does not reflect a DSA effect. The plot of the test results shows that the strength with the temperature tends to deviate near 200 °C (473 K). It might be from a DSA effect.

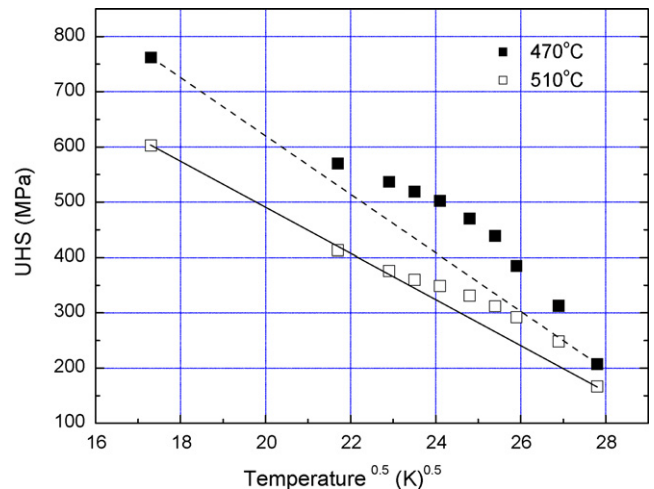


Fig. 5. Effect of test temperature and heat treatment on the tensile stress of Zr–1.5Nb–0.4Sn–0.2Fe cladding tube, strained at $8.33 \times 10^{-5} \text{ s}^{-1}$.

The theoretical strength of the specimen finally heat-treated at 470 °C is 368 MPa at 340 °C near a PWR operation temperature but its UTS was 467 MPa at the same temperature, which is 26.9% higher than the theoretical strength. Whereas, the theoretical strength of the specimen finally heat-treated at 510 °C was 290 MPa at 340 °C but its UTS at the same temperature was 330 MPa which also indicates a 13.8% increase from the theoretical strength. The ductility at a room temperature of the specimen finally heat-treated at 470 and 510 °C was 20.8% and 32.4% but that of the specimen at the 340 °C tensile test was 15.8% and 30.1%, respectively. Thus, it seems that the DSA effect of the specimen made the strength of specimens to be higher but its ductility to be lower. Since the flow stress increases by the repeated lock and release of moving dislocation to approaches the fracture stress, ductility decrease in the DSA range [11]. As the heat treatment temperature increases, indications of DSA such as the increment of strength and the decrement of ductility becomes a little at around 340 °C because lots of the initial dislocation to cause DSA would be annihilated in 510 °C heat-treated condition as mentioned in the previous paragraph.

Fig. 6 indicates the change of yield stress of the specimen with the temperature. The yield stress decreases with temperature as the ultimate tensile stress does. Yield stress is known as the stress above which dislocation starts to move. In the DSA range, the yield stress would not decrease with the change of temperature because a lot of solute atoms adhere to the initial dislocations by a thermal activation and hinder them from moving. With an increase of the heat treatment temperature, the plateau part on the stress–temperature plot, which is supposed to be a DSA region, would shift to a higher temperature. Such an upper shift comes from the fast moving dislocation at the same deformation rate, which is due to the lowering of the dislocation density when the heat treatment temperature becomes higher [10]. Since the diffusivity of the solute atoms needs to be higher to capture the fast dislocation, the DSA region on the plot shifts to a higher temperature if the specimen was heat-treated at a higher temperature. This is closely related to the serration

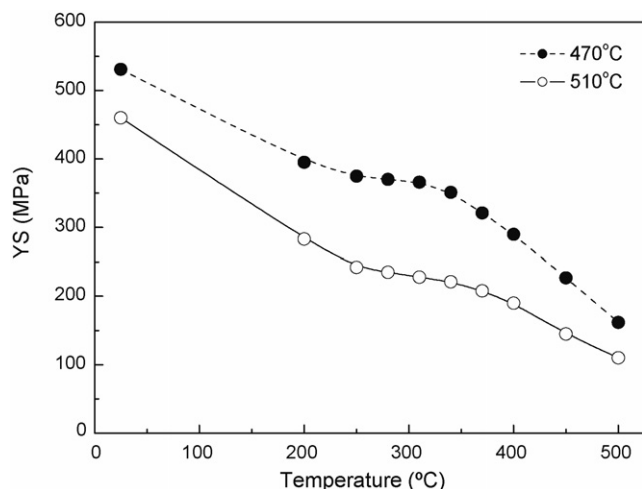


Fig. 6. Effect of test temperature and heat treatment on the yield strength of Zr–1.5Nb–0.4Sn–0.2Fe cladding tube, strained at $8.33 \times 10^{-5} \text{ s}^{-1}$.

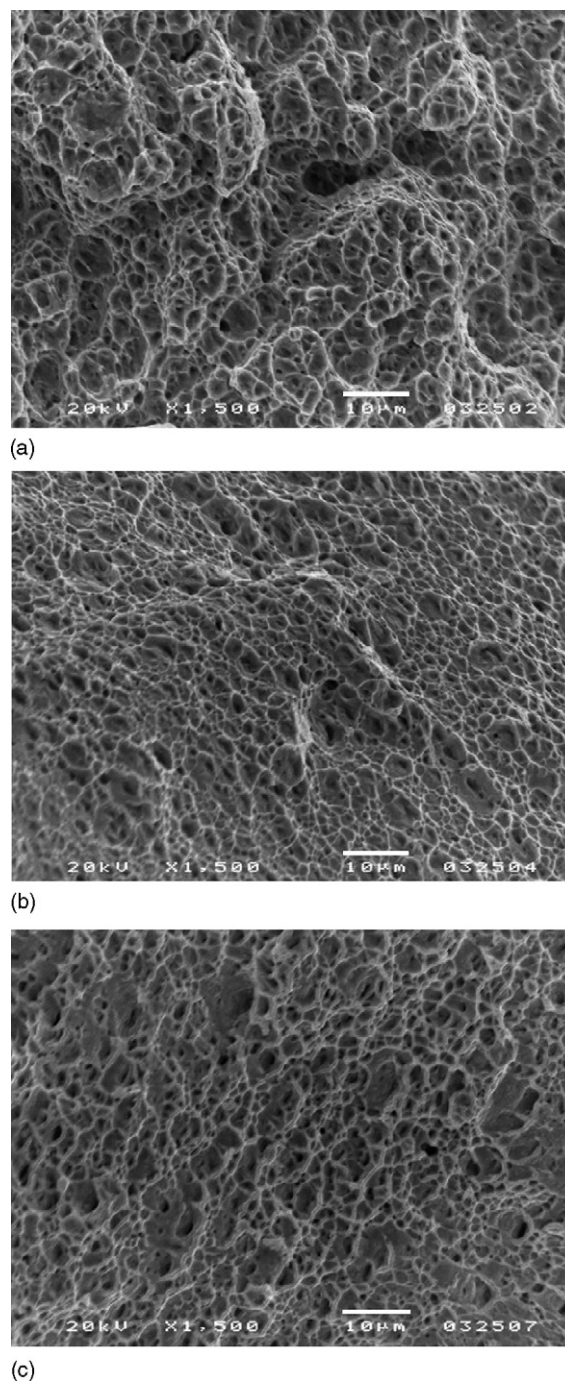


Fig. 7. Fractographs of the stress-relieved Zr–1.5Nb–0.4Sn–0.2Fe cladding tube after tensile test at (a) 200 °C, (b) 340 °C and (c) 500 °C, strained at $8.33 \times 10^{-5} \text{ s}^{-1}$.

shown in Fig. 4 of the specimen heat-treated at 470 °C when it was tested at 340 °C. However, it looks like that the specimen heat-treated at 510 °C would not show a serration at 340 °C because a DSA would occur at a higher temperature. Fig. 7 shows the SEM photographs taken on the fracture surface. Regardless of the test temperature change, the photographs show a dimple surface which is evidence of a ductile fracture. There are no special features on the fracture surface in a DSA temperature range.

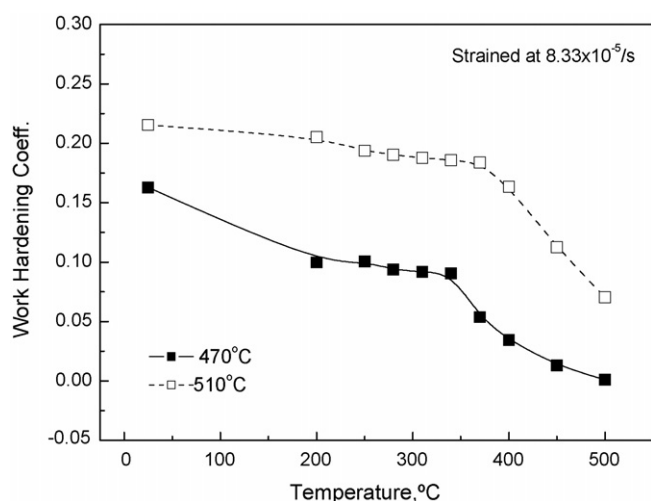


Fig. 8. Effect of the temperature on the work hardening exponent of Zr–1.5Nb–0.4Sn–0.2Fe cladding tube.

3.2. Change of the mechanical properties with the temperature

In general, the portion of the true stress and true strain curve from the onset of yielding to maximum load may be described empirically by the following Hollomon's relationship.

$$\sigma = A\varepsilon^n \quad (1)$$

where σ is the true stress, A the material constant, ε the true plastic strain, and n is the work hardening coefficient.

The relationship can be also described as follows:

$$\log \sigma = n \log \varepsilon + \log A$$

Work hardening coefficient n was calculated by the gradient of the true stress and the true plastic strain on the log scale graph. The coefficient is closely related to the multiplication of dislocation density in the plastic range. Fig. 8 plots the change of the work hardening coefficient with the test temperature. The accumulation of a dislocation or an increase of its density during a plastic deformation brings about an increase of the coefficient. The specimen heat-treated at 510 °C has a higher work hardening coefficient than that heat-treated at 470 °C. The yield stress of the alloy decreases with an increase of the heat treatment temperature which decreases the dislocation density. A greater plastic deformation has to be done in order to gain a faster multiplication of a dislocation which can increase the work hardening coefficient [11]. In general, the shift of the coefficient in a positive direction brings about an intra-granular deformation but a shift to a negative one brings about an inter-granular deformation [3]. In this study, the work hardening coefficient decreased with an increase of the test temperature but it started to increase from 280 °C and showed the highest value at 340 °C. This implies that an intra-granular deformation in the range of a DSA prevails due to the movement of a dislocation through the grains, and the dislocations are fixed to the solute atoms and they make a role for the other dislocation sources for their multiplication to increase the dislocation density and eventually bring about to an increase in the work hardening coefficient [11]. The higher

the test temperature become beyond a DSA interval, the lower the work hardening coefficient becomes regardless of the heat treatment condition of a specimen. This implies that the intra-granular failure of a specimen shifts to an inter-granular failure with an increase of the temperature.

Besides work hardening coefficient, the strain rate sensitivity can be used for an indicator to recognize a DSA. It is defined as a change of the flow stress over a change of the strain rate and can be expressed as follows:

$$m = \frac{d(\ln \sigma)}{d(\ln \dot{\varepsilon})} \approx \frac{\ln \sigma_2 - \ln \sigma_1}{\ln \dot{\varepsilon}_2 - \ln \dot{\varepsilon}_1} \quad (2)$$

where $\dot{\varepsilon}_1 = 8.33 \times 10^{-5} \text{ s}^{-1}$, $\dot{\varepsilon}_2 = 1.67 \times 10^{-2} \text{ s}^{-1}$, σ_1 and σ_2 were yield stresses at 0.2% offset strain when the alloy was strained with $\dot{\varepsilon}_1$ and $\dot{\varepsilon}_2$, respectively.

The flow stress of metals would increase with its strain rate at the same amount of a deformation [12]. But the stress tends to decrease with the strain rate at the same amount of a deformation showing negative strain rate sensitivity during a DSA. The negative strain rate sensitivity comes from the loss of a flow stress when the solute atoms can not capture the moving dislocation any more due to a faster velocity of the dislocation in the material with an increase of the strain rate. Fig. 9 plots the change of the strain rate sensitivity with the temperature. The decrement of the strain rate sensitivity starts at around 200 °C and is of maximum at about 340 °C.

3.3. Review of a DSA mechanism

Zr–1.5Nb–0.4Sn–0.2Fe alloy exhibited the increase of the strength as well as decrease of the strain rate sensitivity in the temperature range of 280–340 °C when it was deformed with the strain rate of 8.33×10^{-5} and $1.67 \times 10^{-2} \text{ s}^{-1}$. To study a DSA mechanism, the activation energy was calculated by the following equation for the given strain rates at a specific temperature considering that a metal has different strain rate response with the temperature when the material have the equal flow stress and

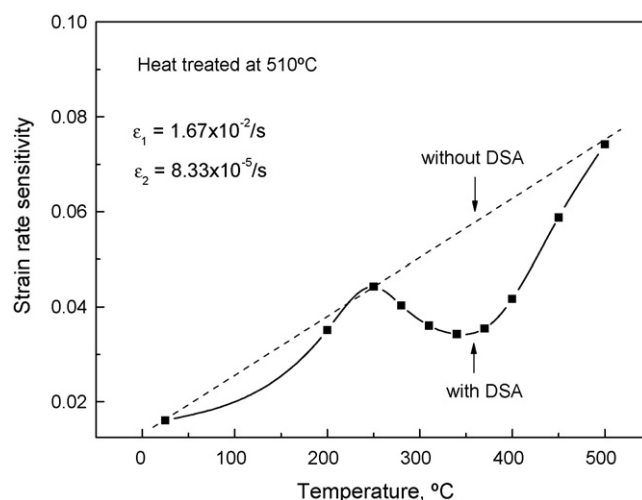


Fig. 9. Temperature effect of Zr–1.5Nb–0.4Sn–0.2Fe cladding tube on the strain rate sensitivity.

strain.

$$\dot{\varepsilon} = A \exp \left(\frac{Q}{RT} \right) \Big|_{\sigma, \varepsilon = \text{const}} \quad (3)$$

The activation energy was about 43.8 kcal/mol at 280 MPa and 0.2% offset strain. It was between the activation energy (30 kcal/mol) of oxygen in a zirconium matrix [13] and that (62 kcal/mol) of a self-diffusion in the zirconium alloy [14]. Considering that a DSA comes from an interaction between moving dislocations and solute atoms, it is recognized that the DSA of Zr–1.5Nb–0.4Sn–0.2Fe alloy could be from a self-diffusion of solute (oxygen) with thermal activation and the self-diffusion is simultaneously supported by the diffusion of vacancies, which might be made around substitutional atoms such as Sn. Although Sn has not a direct effect on the DSA behavior, vacancies around Sn could have an effect on it. Thus, it could be inferred that the solute atom (oxygen) with vacancies around Sn governs the DSA behavior of Zr–1.5Nb–0.4Sn–0.2Fe alloy.

If the deformation of the cladding tube solely depends on the diffusion of the interstitial atoms like oxygen and their locking force for a dislocation, the activation energy of all the deformation with the temperature would almost be equal to that of the solute atoms. If the mechanism is to be governed by only the diffusion of the solute atoms, a deformation can be expressed as follows [15]:

$$\dot{\varepsilon} \approx \frac{4b\rho D}{l} = \frac{4b\rho D_0 \exp(-Q_m/RT)}{l} \quad (4)$$

where b is the Burgers vector of zirconium [16] ($=3.23 \times 10^{-10}$ m), l the effective radius of solute atom (oxygen) [17] ($=0.6 \times 10^{-10}$ m), D the diffusivity of solute atom (oxygen) [18] ($=8 \times 10^{-16}$ m²/s at 340 °C), and ρ is the dislocation density of the matrix 10^8 m/m³ for stress-relieved condition, 10^5 m/m³ for partially recrystallized condition [19]

According to the above relationship, the strain rate for a DSA of the stress-relieved zirconium alloy is calculated as follows:

$$\begin{aligned} \dot{\varepsilon} &\approx \frac{4 \times 3.23 \times 10^{-10} \text{ m} \times 10^8 \text{ m/m}^3 \times 8 \times 10^{-16} \text{ m}^2/\text{s}}{0.6 \times 10^{-10} \text{ m}} \\ &= 1.72 \times 10^{-6} \text{ s}^{-1} \end{aligned} \quad (5)$$

The DSA of the Zr–1.5Nb–0.4Sn–0.2Fe alloy activates at the strain rate of about 10^{-6} s^{−1}.

It suggests that the solute atoms can capture the moving dislocation at a low strain rate of about 10^{-6} s^{−1} if the alloy is deformed by only the diffusion of the solute atoms. Since a DSA of the alloy was observed at the rapid strain rate of 10^{-2} s^{−1} as well as at the slow strain rate of 10^{-5} s^{−1}, it seems that the intervention of other mechanism than that by the interstitial atoms into the deformation causes an increase in the strain rate to bring about a DSA. Thus, it is imaginable that the movement of vacancy around substitutional atom like Sn in the alloy encourages the diffusion of the solute atoms to increase the strain rate for an initiation of DSA. If the diffusion of the interstitial atoms is affected the migration of vacancies around a

substitutional atom, the deformation can be expressed as follows [20]:

$$\dot{\varepsilon} \approx \frac{4b\rho C_v D}{l} = \frac{4b\rho C_v D_0 \exp(-Q_m/RT)}{l} \quad (6)$$

where C_v is the vacancy density. C_v increases with an increase of the dislocation density as a plastic deformation becomes greater. The vacancies from a plastic deformation can be a space for solute atoms having a high mobility with the help of a thermal activation to capture the fast moving dislocation. Eventually, a DSA of the alloy could be affected at the fast strain rate of 10^{-2} s^{−1}.

4. Conclusion

To investigate the mechanical properties and to study a DSA phenomenon of the Zr–1.5Nb–0.4Sn–0.2Fe alloy for a nuclear cladding tube, tensile tests of the tube specimens were carried out in the range of a room temperature to 500 °C with the strain rate of 8.33×10^{-5} and 1.67×10^{-2} s^{−1}. The test results can be summarized as follows:

- (1) The followings were observed above 200 °C on the Zr–1.5Nb–0.4Sn–0.2Fe alloy that had been given a final heat treatment both stress-relieved at 470 °C and partially recrystallized at 510 °C. (a) A serration due to a DSA, an increase of the tensile stress with a decrease of the ductility, (b) an increase of the work hardening coefficient, and (c) a decrease of the strain rate sensitivity.
- (2) If the alloy was finally heat-treated at a higher temperature, the temperature region to induce a DSA shifted to a higher temperature side due to a decrease of the initial dislocation density.
- (3) The activation energy for a deformation and the model of an atomic diffusion was reviewed to study the mechanism of a DSA on the alloy. The main mechanism to initiate a DSA in the Zr–1.5Nb–0.4Sn–0.2Fe alloy is not only the diffusion of the interstitial atom such as oxygen alone but also its acceleration caused by the migration of the vacancies around the substitutional atoms such as Sn. Thus, it would be possible to bring about a DSA on the alloy even at a strain rate of 10^{-2} s^{−1}.

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